THE NOVA RESEARCH CURRICULUM

Biology

Grade 8 · Term 3

Student edition

Outcome: BI-8.3 carbon and oxygen cycles, food webs, global warming and uses of biotechnology.

Lesson notes

- **Cycles (Weeks 1–4):** carbon cycle: plants take in carbon dioxide in photosynthesis; respiration, burning fuels and decomposers release it. Oxygen cycle: photosynthesis releases oxygen; respiration and burning use it.

  ▶ **Try it on NovaSchool:** The carbon cycle — open **Simulations** on your dashboard.

- **Food webs and global warming (Weeks 5–8):** producers, primary and secondary consumers, decomposers; only a small part of the energy passes to the next level. Greenhouse gases (carbon dioxide, methane) trap heat; deforestation and burning fuels add to them. Effects in Pakistan: melting glaciers, heatwaves, floods. Solutions: trees, renewable energy, saving electricity.
- **Biotechnology (Weeks 9–12):** using living things to make useful products: yeast in bread, bacteria in yoghurt, bacteria engineered to make insulin, pest-resistant (Bt) cotton grown in Pakistan. Discuss benefits and concerns.

Workbook pages

Page	Task
1	Draw the carbon cycle with arrows.
2	Food web of a Pakistani wetland.
3	Greenhouse-effect diagram and my action plan.
4	Biotechnology products table.

Question bank

#	Outcome	Question	Your answer
1	BI-8.3	Which process removes carbon dioxide from the air?	respiration / burning / photosynthesis / decay
2	BI-8.3	Organisms that break down dead material are …	producers / decomposers / consumers / predators
3	BI-8.3	Which is a greenhouse gas?	oxygen / nitrogen / methane / argon
4	BI-8.3	Cutting down forests increases carbon dioxide in the air. True or false.	True / False

#	Outcome	Question	Your answer
5	BI-8.3	Insulin for diabetes can now be made by ...	plants / engineered bacteria / rocks / yeast only
6	BI-8.3	Yoghurt is made using ...	yeast / bacteria / viruses / fungi only
7	BI-8.3	Give two effects of global warming on Pakistan.	_____
8	BI-8.3	All of the energy at one level passes to the next level. True or false.	True / False